

Notebook LLM Lite

Multi-Document Research AI with Semantic Search

Core Objective LITE

Build a "Second Brain" for studying. Upload multiple PDFs and ask questions across all documents using RAG with PostgreSQL.

Key Features:

- Upload & manage multiple documents
- Query across all uploaded documents
- Vue.js for reactive front-end interface

Tech Stack

Vue.js 3 Groq API Docling

PostgreSQL FastAPI

Why Vue.js? Reactive, component-based, and perfect for managing multiple documents and streaming data responses.

Meaning is Geometry: Cosine Similarity

Convert sentences into vectors (384 numbers). Find the "angle" between the User's Question and Document Chunks across all uploaded files.

$$\text{similarity} = (A \cdot B) / (||A|| \times ||B||)$$

```
# Postgres pgvector similarity search
CREATE EXTENSION vector;
CREATE TABLE documents (
  id SERIAL PRIMARY KEY,
  filename VARCHAR(255),
  content TEXT,
  embedding vector(384)
);

SELECT content, filename FROM documents
ORDER BY embedding <=> '[0.1, 0.2, ... ]'::vector
LIMIT 5;
```

Vue.js Multi-Document Components

App.vue (Main Component)

Parent

Manages multiple documents and global application state

```
import { ref } from 'vue'
import DocumentManager from './components/DocumentManager.vue'
import ChatInterface from './components/ChatInterface.vue'

// State for multiple documents
const documents = ref([]) // Array of uploaded files
```

DocumentManager.vue

Child

Handles uploading multiple PDFs and displays document list

```
// Handle multiple file upload
const uploadFiles = async (files) => {
  for (const file of files) {
    const formData = new FormData()
    formData.append('file', file)
    await api.uploadDocument(formData)
    documents.value.push(file)
  }
}
```

ChatInterface.vue

Child

Queries across all documents and displays citations from relevant sources

```
// Query all documents
async function sendMessage(question) {
  const response = await fetch('/api/chat', {
    method: 'POST',
    body: JSON.stringify({ question, allDocs: true })
  })
  // Stream response with citations from multiple docs
}
```

Multi-Document Interface Preview

Vector-Brain Notebook (Vue.js 3)

Uploaded Documents

How do rockets work?

Physics_Textbook.pdf
245 pages • 4.2 MB

Chemistry_Basics.pdf
180 pages • 3.1 MB

Research_Paper.pdf
15 pages • 0.8 MB

Rockets work by expelling exhaust in the opposite direction of travel, creating thrust through Newton's third law of motion.

Source: Physics_Textbook.pdf, p. 45

What are the chemical components of rocket fuel?

Common rocket fuels include liquid hydrogen with liquid oxygen, kerosene with liquid oxygen, and solid propellants like ammonium perchlorate composite.

Ask a question across all documents...

Send

Definition of Done



Multiple
PDFs



Query All
Documents



Answer < 1s
Latency



PostgreSQL
Database



Vue.js
Frontend